CORE PURE / FURTHER CALCULUS

Integration by Substitution

Worked solutions 15 questions · 113 marks

Questions

Question 1	*7 marks*	2
Question 2	*8 marks*	4
Question 3	*8 marks*	6
Question 4	*6 marks*	8
Question 5	*8 marks*	10
Question 6	*6 marks*	12
Question 7	*7 marks*	14
Question 8	*8 marks*	16
Question 9	*7 marks*	18
Question 10	*7 marks*	20
Question 11	*9 marks*	22
Question 12	*8 marks*	24
Question 13	*9 marks*	26
Question 14	*7 marks*	28
Question 15	*8 marks*	29

1. Show that

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}}\,\mathrm{d}x = \frac{16}{15}(7-4\sqrt{2}) \qquad \textbf{[7]}$$

Solution

Let

$$I = \int_1^3 \frac{(x-1)^2}{\sqrt{x+1}}\,\mathrm{d}x$$

Use the substitution

$$u = x+1$$

so that

$$x = u-1, \qquad x-1 = u-2, \qquad \mathrm{d}x = \mathrm{d}u$$

The limits also change:

$$x = 1 \Rightarrow u = 2, \qquad x = 3 \Rightarrow u = 4$$

So

$$\begin{aligned} I &= \int_2^4 \frac{(u-2)^2}{\sqrt{u}}\,\mathrm{d}u \\ &= \int_2^4 (u-2)^2 u^{-1/2}\,\mathrm{d}u \\ &= \int_2^4 (u^2-4u+4)u^{-1/2}\,\mathrm{d}u \\ &= \int_2^4 \left(u^{3/2}-4u^{1/2}+4u^{-1/2}\right)\mathrm{d}u \end{aligned}$$

Integrating term by term,

$$\begin{aligned} I &= \left[\frac{2}{5}u^{5/2} - 4\cdot\frac{2}{3}u^{3/2} + 4\cdot 2u^{1/2}\right]_2^4 \\ &= \left[\frac{2}{5}u^{5/2} - \frac{8}{3}u^{3/2} + 8u^{1/2}\right]_2^4 \end{aligned}$$

Now evaluate at the limits.
When $u = 4$,

$$4^{1/2} = 2, \qquad 4^{3/2} = 8, \qquad 4^{5/2} = 32$$

so

$$\frac{2}{5}(32) - \frac{8}{3}(8) + 8(2) = \frac{64}{5} - \frac{64}{3} + 16$$

When $u = 2$,

$$2^{1/2} = \sqrt{2}, \qquad 2^{3/2} = 2\sqrt{2}, \qquad 2^{5/2} = 4\sqrt{2}$$

so

$$\frac{2}{5}(4\sqrt{2}) - \frac{8}{3}(2\sqrt{2}) + 8\sqrt{2} = \frac{8\sqrt{2}}{5} - \frac{16\sqrt{2}}{3} + 8\sqrt{2}$$

Hence

$$\begin{aligned} I &= \left(\frac{64}{5} - \frac{64}{3} + 16\right) - \left(\frac{8\sqrt{2}}{5} - \frac{16\sqrt{2}}{3} + 8\sqrt{2}\right) \\ &= \frac{192-320+240}{15} - \frac{24\sqrt{2}-80\sqrt{2}+120\sqrt{2}}{15} \\ &= \frac{112}{15} - \frac{64\sqrt{2}}{15} \\ &= \frac{16}{15}(7-4\sqrt{2}) \end{aligned}$$

Therefore

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}}\,\mathrm{d}x = \frac{16}{15}(7-4\sqrt{2})$$

as required.

2. **(a)** Use the substitution $u = 1 + x^2$ to show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_p^q \frac{(u-1)^4}{2u}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = A + B\ln 5$$

where A and B are constants to be found. **[5]**

Solution

(a) Let

$$u = 1 + x^2$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x \qquad \Rightarrow \qquad \mathrm{d}u = 2x\,\mathrm{d}x$$

Also,

$$x^9\,\mathrm{d}x = x^8(x\,\mathrm{d}x) = (x^2)^4 \cdot \frac{\mathrm{d}u}{2} = (u-1)^4 \cdot \frac{\mathrm{d}u}{2}$$

and

$$1 + x^2 = u$$

So

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x &= \int \frac{x^9\,\mathrm{d}x}{1+x^2} \\ &= \int \frac{(u-1)^4/2}{u}\,\mathrm{d}u\end{aligned}$$

Now change the limits:

$$x = 0 \Rightarrow u = 1 + 0^2 = 1$$
$$x = 2 \Rightarrow u = 1 + 2^2 = 5$$

Hence

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_1^5 \frac{(u-1)^4}{2u}\,\mathrm{d}u$$

Therefore,

$$p = 1, \qquad q = 5$$

(b) From part (a),

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_1^5 \frac{(u-1)^4}{2u}\,\mathrm{d}u$$

Expand:

$$(u-1)^4 = u^4 - 4u^3 + 6u^2 - 4u + 1$$
$$\frac{(u-1)^4}{u} = u^3 - 4u^2 + 6u - 4 + \frac{1}{u}$$

So

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x &= \frac{1}{2}\int_1^5 \left(u^3 - 4u^2 + 6u - 4 + \frac{1}{u}\right)\mathrm{d}u \\ &= \left[\frac{u^4}{8} - \frac{2u^3}{3} + \frac{3u^2}{2} - 2u + \frac{1}{2}\ln u\right]_1^5\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x &= \left(\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 + \frac{1}{2}\ln 5\right) - \left(\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 + \frac{1}{2}\ln 1\right)\\ &= \left(\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 + \frac{1}{2}\ln 5\right) - \left(\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2\right)\end{aligned}$$

Since

$$\frac{625}{8} - \frac{250}{3} + \frac{75}{2} - 10 = \frac{535}{24}$$

and

$$\frac{1}{8} - \frac{2}{3} + \frac{3}{2} - 2 = -\frac{25}{24}$$

we get

$$\begin{aligned}\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x &= \frac{535}{24} + \frac{1}{2}\ln 5 - \left(-\frac{25}{24}\right)\\ &= \frac{560}{24} + \frac{1}{2}\ln 5\\ &= \frac{70}{3} + \frac{1}{2}\ln 5\end{aligned}$$

Hence

$$A = \frac{70}{3}, \qquad B = \frac{1}{2}$$

3. Using calculus, find the exact values of

(i)

$$\int_{-2}^{0} \frac{1}{x^2+4x+8}\,\mathrm{d}x \qquad \textbf{[3]}$$

(ii)

$$\int_{0}^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}}\,\mathrm{d}x \qquad \textbf{[5]}$$

Solution

(i) Complete the square in the denominator:

$$x^2+4x+8 = (x+2)^2+4$$

Use the standard result

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+c$$

with $a=2$, and with x replaced by $x+2$:

$$\begin{aligned}\int_{-2}^{0} \frac{1}{x^2+4x+8}\,\mathrm{d}x &= \int_{-2}^{0} \frac{1}{(x+2)^2+4}\,\mathrm{d}x \\ &= \left[\frac{1}{2}\arctan\left(\frac{x+2}{2}\right)\right]_{-2}^{0} \\ &= \frac{1}{2}\left(\arctan 1 - \arctan 0\right) \\ &= \frac{1}{2}\left(\frac{\pi}{4}-0\right) \\ &= \frac{\pi}{8}\end{aligned}$$

(ii) Use the substitution

$$x = 2\sin\theta$$

Then

$$\mathrm{d}x = 2\cos\theta\,\mathrm{d}\theta$$

and

$$\sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = 2\cos\theta$$

Since $0 \le x \le \sqrt{3}$, we have $0 \le \theta \le \frac{\pi}{3}$, so $\cos\theta > 0$.

Change the limits:

$$x = 0 \Rightarrow \theta = 0, \qquad x = \sqrt{3} \Rightarrow 2\sin\theta = \sqrt{3} \Rightarrow \theta = \frac{\pi}{3}$$

Now substitute:

$$\begin{aligned}\int_{0}^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}}\,\mathrm{d}x &= \int_{0}^{\pi/3} \frac{(2\sin\theta)^2}{2\cos\theta}\cdot 2\cos\theta\,\mathrm{d}\theta \\ &= \int_{0}^{\pi/3} 4\sin^2\theta\,\mathrm{d}\theta\end{aligned}$$

Use

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

So

$$\begin{aligned}\int_0^{\pi/3} 4\sin^2\theta\,\mathrm{d}\theta &= 4\int_0^{\pi/3}\frac{1-\cos 2\theta}{2}\,\mathrm{d}\theta \\ &= 2\int_0^{\pi/3}(1-\cos 2\theta)\,\mathrm{d}\theta \\ &= 2\left[\theta-\frac{1}{2}\sin 2\theta\right]_0^{\pi/3}\end{aligned}$$

Now evaluate:

$$\begin{aligned}2\left[\theta-\frac{1}{2}\sin 2\theta\right]_0^{\pi/3} &= 2\left(\frac{\pi}{3}-\frac{1}{2}\sin\frac{2\pi}{3}\right) \\ &= 2\left(\frac{\pi}{3}-\frac{1}{2}\cdot\frac{\sqrt{3}}{2}\right) \\ &= \frac{2\pi}{3}-\frac{\sqrt{3}}{2}\end{aligned}$$

4. Show that

$$\int \frac{\sqrt{x^2-16}}{x^2}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c \qquad \textbf{[6]}$$

Solution

Let

$$I = \int \frac{\sqrt{x^2-16}}{x^2}\,\mathrm{d}x$$

Use the substitution

$$x = 4\cosh u$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 4\sinh u \qquad \Rightarrow \qquad \mathrm{d}x = 4\sinh u\,\mathrm{d}u$$

Also,

$$\begin{aligned}\sqrt{x^2-16} &= \sqrt{16\cosh^2 u - 16}\\ &= 4\sqrt{\cosh^2 u - 1}\\ &= 4\sqrt{\sinh^2 u}\\ &= 4\sinh u\end{aligned}$$

since $u > 0$ gives $\sinh u > 0$.
Substituting into the integral,

$$\begin{aligned}I &= \int \frac{4\sinh u}{(4\cosh u)^2}\cdot 4\sinh u\,\mathrm{d}u\\ &= \int \frac{16\sinh^2 u}{16\cosh^2 u}\,\mathrm{d}u\\ &= \int \frac{\sinh^2 u}{\cosh^2 u}\,\mathrm{d}u\\ &= \int \tanh^2 u\,\mathrm{d}u\end{aligned}$$

Now use

$$\tanh^2 u = 1 - \operatorname{sech}^2 u$$

so

$$\begin{aligned}I &= \int \left(1 - \operatorname{sech}^2 u\right)\,\mathrm{d}u\\ &= u - \tanh u + c\end{aligned}$$

Now write this in terms of x.
From $x = 4\cosh u$,

$$\cosh u = \frac{x}{4}$$

so, since $u > 0$,

$$u = \operatorname{arcosh}\left(\frac{x}{4}\right)$$

Also,

$$\begin{aligned}\tanh u &= \frac{\sinh u}{\cosh u}\\ &= \frac{\sqrt{x^2-16}/4}{x/4}\\ &= \frac{\sqrt{x^2-16}}{x}\end{aligned}$$

Therefore,

$$I = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c$$

Hence,

$$\int \frac{\sqrt{x^2-16}}{x^2}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c$$

5. Using calculus, find the exact values of

(i)
$$\int_1^2 \frac{1}{x^2+3x+2}\,\mathrm{d}x \qquad \textbf{[3]}$$

(ii)
$$\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x \qquad \textbf{[5]}$$

Solution

(i) First factorise the denominator:
$$x^2+3x+2=(x+1)(x+2)$$
Use partial fractions:
$$\frac{1}{(x+1)(x+2)}=\frac{A}{x+1}+\frac{B}{x+2}$$
So
$$1=A(x+2)+B(x+1)$$
Putting $x=-1$ gives $A=1$, and putting $x=-2$ gives $B=-1$. Hence
$$\frac{1}{x^2+3x+2}=\frac{1}{x+1}-\frac{1}{x+2}$$
Therefore
$$\begin{aligned}\int_1^2 \frac{1}{x^2+3x+2}\,\mathrm{d}x &= \int_1^2 \left(\frac{1}{x+1}-\frac{1}{x+2}\right)\mathrm{d}x\\ &= [\ln(x+1)-\ln(x+2)]_1^2\\ &= (\ln 3-\ln 4)-(\ln 2-\ln 3)\\ &= 2\ln 3-\ln 4-\ln 2\\ &= \ln\left(\frac{9}{8}\right)\end{aligned}$$

(ii) Use the substitution
$$x=2\sin\theta$$
Then
$$\frac{\mathrm{d}x}{\mathrm{d}\theta}=2\cos\theta \qquad \Rightarrow \qquad \mathrm{d}x=2\cos\theta\,\mathrm{d}\theta$$
Also,
$$\sqrt{4-x^2}=\sqrt{4-4\sin^2\theta}=2\cos\theta$$
since on the interval we use, θ lies between $\frac{\pi}{6}$ and $\frac{\pi}{3}$, so $\cos\theta>0$.
Now change the limits.
When $x=1$,
$$1=2\sin\theta \quad\Rightarrow\quad \sin\theta=\frac{1}{2} \quad\Rightarrow\quad \theta=\frac{\pi}{6}$$
When $x=\sqrt{3}$,
$$\sqrt{3}=2\sin\theta \quad\Rightarrow\quad \sin\theta=\frac{\sqrt{3}}{2} \quad\Rightarrow\quad \theta=\frac{\pi}{3}$$

So

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x &= \int_{\pi/6}^{\pi/3} \frac{2\cos\theta}{(2\sin\theta)^2}\cdot 2\cos\theta\,\mathrm{d}\theta \\ &= \int_{\pi/6}^{\pi/3} \frac{4\cos^2\theta}{4\sin^2\theta}\,\mathrm{d}\theta \\ &= \int_{\pi/6}^{\pi/3} \cot^2\theta\,\mathrm{d}\theta\end{aligned}$$

Use the identity

$$\cot^2\theta = \operatorname{cosec}^2\theta - 1$$

Then

$$\begin{aligned}\int_{\pi/6}^{\pi/3} \cot^2\theta\,\mathrm{d}\theta &= \int_{\pi/6}^{\pi/3} (\operatorname{cosec}^2\theta - 1)\,\mathrm{d}\theta \\ &= [-\cot\theta - \theta]_{\pi/6}^{\pi/3}\end{aligned}$$

Now evaluate:

$$\begin{aligned}[-\cot\theta - \theta]_{\pi/6}^{\pi/3} &= \left(-\cot\frac{\pi}{3} - \frac{\pi}{3}\right) - \left(-\cot\frac{\pi}{6} - \frac{\pi}{6}\right) \\ &= \left(-\frac{1}{\sqrt{3}} - \frac{\pi}{3}\right) - \left(-\sqrt{3} - \frac{\pi}{6}\right) \\ &= -\frac{\sqrt{3}}{3} - \frac{\pi}{3} + \sqrt{3} + \frac{\pi}{6} \\ &= \frac{2\sqrt{3}}{3} - \frac{\pi}{6}\end{aligned}$$

6. Show that

$$\int \frac{x^2}{\sqrt{x^2+16}}\,\mathrm{d}x = \frac{1}{2}x\sqrt{x^2+16} - 8\,\mathrm{arsinh}\left(\frac{x}{4}\right) + c \qquad \textbf{[6]}$$

Solution

Let

$$x = 4\sinh u$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 4\cosh u \qquad \Rightarrow \qquad \mathrm{d}x = 4\cosh u\,\mathrm{d}u$$

Also,

$$\begin{aligned}\sqrt{x^2+16} &= \sqrt{16\sinh^2 u + 16}\\ &= \sqrt{16(\sinh^2 u + 1)}\\ &= \sqrt{16\cosh^2 u}\\ &= 4\cosh u\end{aligned}$$

since $\cosh u > 0$ for real u.
Now substitute into the integral:

$$\begin{aligned}\int \frac{x^2}{\sqrt{x^2+16}}\,\mathrm{d}x &= \int \frac{(4\sinh u)^2}{4\cosh u}\cdot 4\cosh u\,\mathrm{d}u\\ &= \int 16\sinh^2 u\,\mathrm{d}u\\ &= 16\int \sinh^2 u\,\mathrm{d}u\end{aligned}$$

Use the identity

$$\sinh^2 u = \frac{\cosh 2u - 1}{2}$$

So

$$\begin{aligned}16\int \sinh^2 u\,\mathrm{d}u &= 16\int \frac{\cosh 2u - 1}{2}\,\mathrm{d}u\\ &= 8\int (\cosh 2u - 1)\,\mathrm{d}u\\ &= 8\left(\frac{1}{2}\sinh 2u - u\right) + c\\ &= 4\sinh 2u - 8u + c\end{aligned}$$

Now use

$$\sinh 2u = 2\sinh u\cosh u$$

giving

$$4\sinh 2u - 8u = 8\sinh u\cosh u - 8u$$

Now convert back to x.
From $x = 4\sinh u$,

$$\sinh u = \frac{x}{4}$$

and since $\sqrt{x^2+16} = 4\cosh u$,

$$\cosh u = \frac{\sqrt{x^2+16}}{4}$$

Also,

$$u = \operatorname{arsinh}\left(\frac{x}{4}\right)$$

Therefore

$$\begin{aligned} 8\sinh u\cosh u - 8u + c &= 8\left(\frac{x}{4}\right)\left(\frac{\sqrt{x^2+16}}{4}\right) - 8\operatorname{arsinh}\left(\frac{x}{4}\right) + c \\ &= \frac{1}{2}x\sqrt{x^2+16} - 8\operatorname{arsinh}\left(\frac{x}{4}\right) + c \end{aligned}$$

Hence

$$\int \frac{x^2}{\sqrt{x^2+16}}\,\mathrm{d}x = \frac{1}{2}x\sqrt{x^2+16} - 8\operatorname{arsinh}\left(\frac{x}{4}\right) + c$$

7. Show that

$$\int_0^{\ln 2} \frac{e^{3x}+2e^{2x}}{\sqrt{e^x+4}}\,\mathrm{d}x = \frac{32\sqrt{6}-30\sqrt{5}}{5} \qquad [7]$$

Solution

Let

$$I = \int_0^{\ln 2} \frac{e^{3x}+2e^{2x}}{\sqrt{e^x+4}}\,\mathrm{d}x$$

Use the substitution $u = e^x$. Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = e^x = u \qquad \Rightarrow \qquad \mathrm{d}u = e^x\,\mathrm{d}x, \quad \mathrm{d}x = \frac{\mathrm{d}u}{u}$$

Also, the limits change as follows:

$$x = 0 \Rightarrow u = 1, \qquad x = \ln 2 \Rightarrow u = 2$$

So

$$\begin{aligned} I &= \int_1^2 \frac{u^3+2u^2}{\sqrt{u+4}} \cdot \frac{1}{u}\,\mathrm{d}u \\ &= \int_1^2 \frac{u^2+2u}{\sqrt{u+4}}\,\mathrm{d}u \end{aligned}$$

Now use a second substitution:

$$v = u+4 \qquad \Rightarrow \qquad \mathrm{d}v = \mathrm{d}u, \quad u = v-4$$

The new limits are

$$u = 1 \Rightarrow v = 5, \qquad u = 2 \Rightarrow v = 6$$

Therefore

$$\begin{aligned} I &= \int_5^6 \frac{(v-4)^2+2(v-4)}{\sqrt{v}}\,\mathrm{d}v \\ &= \int_5^6 \frac{v^2-8v+16+2v-8}{\sqrt{v}}\,\mathrm{d}v \\ &= \int_5^6 \frac{v^2-6v+8}{\sqrt{v}}\,\mathrm{d}v \\ &= \int_5^6 \left(v^{3/2}-6v^{1/2}+8v^{-1/2}\right)\mathrm{d}v \end{aligned}$$

Integrate term by term:

$$I = \left[\frac{2}{5}v^{5/2}-4v^{3/2}+16v^{1/2}\right]_5^6$$

Now substitute the limits:

$$\begin{aligned} I &= \left(\frac{2}{5}\cdot 6^{5/2}-4\cdot 6^{3/2}+16\cdot 6^{1/2}\right)-\left(\frac{2}{5}\cdot 5^{5/2}-4\cdot 5^{3/2}+16\cdot 5^{1/2}\right) \\ &= \left(\frac{2}{5}\cdot 36\sqrt{6}-4\cdot 6\sqrt{6}+16\sqrt{6}\right)-\left(\frac{2}{5}\cdot 25\sqrt{5}-4\cdot 5\sqrt{5}+16\sqrt{5}\right) \\ &= \left(\frac{72}{5}\sqrt{6}-24\sqrt{6}+16\sqrt{6}\right)-\left(10\sqrt{5}-20\sqrt{5}+16\sqrt{5}\right) \\ &= \left(\frac{72}{5}-8\right)\sqrt{6}-6\sqrt{5} \\ &= \frac{32}{5}\sqrt{6}-6\sqrt{5} \\ &= \frac{32\sqrt{6}-30\sqrt{5}}{5} \end{aligned}$$

Hence

$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}}\,\mathrm{d}x = \frac{32\sqrt{6} - 30\sqrt{5}}{5}$$

as required.

8. Using the substitution $u = \frac{2}{3}\tanh x$, evaluate

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x \qquad \textbf{[8]}$$

Solution

Let

$$I = \int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x$$

Use the given substitution

$$u = \frac{2}{3}\tanh x$$

Differentiate:

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2}{3}\operatorname{sech}^2 x$$

so

$$\mathrm{d}u = \frac{2}{3}\operatorname{sech}^2 x\,\mathrm{d}x$$

and hence

$$\operatorname{sech}^2 x\,\mathrm{d}x = \frac{3}{2}\,\mathrm{d}u$$

Also, from $u = \frac{2}{3}\tanh x$,

$$\tanh x = \frac{3}{2}u$$

so the expression inside the square root becomes

$$\begin{aligned} 9 - 4\tanh^2 x &= 9 - 4\left(\frac{3}{2}u\right)^2 \\ &= 9 - 4\cdot\frac{9}{4}u^2 \\ &= 9 - 9u^2 \\ &= 9(1 - u^2) \end{aligned}$$

Therefore,

$$\sqrt{9 - 4\tanh^2 x} = 3\sqrt{1 - u^2}$$

Now change the limits.
When $x = 0$,

$$u = \frac{2}{3}\tanh 0 = 0$$

When $x = 1$,

$$u = \frac{2}{3}\tanh 1$$

Substituting everything into the integral:

$$\begin{aligned} I &= \int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x \\ &= \int_0^{\frac{2}{3}\tanh 1} \frac{\frac{3}{2}\,\mathrm{d}u}{3\sqrt{1 - u^2}} \\ &= \frac{1}{2}\int_0^{\frac{2}{3}\tanh 1} \frac{1}{\sqrt{1 - u^2}}\,\mathrm{d}u \end{aligned}$$

Using

$$\int \frac{1}{\sqrt{1 - u^2}}\,\mathrm{d}u = \arcsin u$$

we get

$$\begin{aligned} I &= \frac{1}{2}\left[\arcsin u\right]_0^{\frac{2}{3}\tanh 1} \\ &= \frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right) - \frac{1}{2}\arcsin(0) \\ &= \frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right) \end{aligned}$$

So the exact value is

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x = \frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right)$$

Numerically,

$$\frac{1}{2}\arcsin\left(\frac{2}{3}\tanh 1\right) \approx 0.266$$

9. **(a)** Use the substitution $u = \sqrt{2x+1}$ to show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = \int_p^q \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = A - B\ln 2$$

where A and B are constants to be found. **[4]**

Solution

(a) Let

$$u = \sqrt{2x+1}$$

Then

$$u^2 = 2x+1 \qquad \Rightarrow \qquad x = \frac{u^2-1}{2}$$

Differentiate $u^2 = 2x+1$ with respect to x:

$$2u\frac{\mathrm{d}u}{\mathrm{d}x} = 2$$

so

$$\frac{\mathrm{d}x}{\mathrm{d}u} = u \qquad \Rightarrow \qquad \mathrm{d}x = u\,\mathrm{d}u$$

Also,

$$\sqrt{2x+1} = u$$

Now change the limits.
When $x = 0$,

$$u = \sqrt{2(0)+1} = 1$$

When $x = \frac{15}{2}$,

$$u = \sqrt{2\cdot\frac{15}{2}+1} = \sqrt{16} = 4$$

Therefore

$$\begin{aligned}\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x &= \int_1^4 \frac{\frac{u^2-1}{2}}{2+u}\cdot u\,\mathrm{d}u \\ &= \int_1^4 \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u\end{aligned}$$

Hence $p = 1$ and $q = 4$.

(b) Using part (a),

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = \int_1^4 \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u$$

First simplify the integrand. Since

$$u(u^2-1) = u^3 - u$$

divide $u^3 - u$ by $u+2$:

$$u^3 - u = (u+2)(u^2-2u+3) - 6$$

So

$$\frac{u(u^2-1)}{2(u+2)} = \frac{1}{2}\left(u^2 - 2u + 3 - \frac{6}{u+2}\right)$$

Hence

$$\begin{aligned}\int_1^4 \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u &= \frac{1}{2}\int_1^4 \left(u^2 - 2u + 3 - \frac{6}{u+2}\right)\mathrm{d}u \\ &= \frac{1}{2}\left[\frac{u^3}{3} - u^2 + 3u - 6\ln(u+2)\right]_1^4\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}&= \frac{1}{2}\left[\left(\frac{4^3}{3} - 4^2 + 3\cdot 4 - 6\ln 6\right) - \left(\frac{1^3}{3} - 1^2 + 3\cdot 1 - 6\ln 3\right)\right] \\ &= \frac{1}{2}\left[\left(\frac{64}{3} - 16 + 12 - 6\ln 6\right) - \left(\frac{1}{3} - 1 + 3 - 6\ln 3\right)\right] \\ &= \frac{1}{2}\left[\left(\frac{52}{3} - 6\ln 6\right) - \left(\frac{7}{3} - 6\ln 3\right)\right] \\ &= \frac{1}{2}\left[\frac{45}{3} - 6(\ln 6 - \ln 3)\right] \\ &= \frac{1}{2}\left[15 - 6\ln\left(\frac{6}{3}\right)\right] \\ &= \frac{1}{2}(15 - 6\ln 2) \\ &= \frac{15}{2} - 3\ln 2\end{aligned}$$

Therefore

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = \frac{15}{2} - 3\ln 2$$

So

$$A = \frac{15}{2}, \qquad B = 3$$

10. Use an appropriate substitution to find the exact value of

$$\int_0^{\pi/6} \frac{\cos x \cos 2x}{(1+\sin x)^{3/2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\sqrt{6}$, where a and b are rational numbers to be determined. **[7]**

Solution

Let

$$I = \int_0^{\frac{\pi}{6}} \frac{\cos x \cos 2x}{(1+\sin x)^{3/2}}\,\mathrm{d}x$$

Use the substitution

$$u = 1 + \sin x$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \qquad \Rightarrow \qquad \mathrm{d}u = \cos x\,\mathrm{d}x$$

The limits change as follows:

$$x = 0 \Rightarrow u = 1 + \sin 0 = 1$$
$$x = \frac{\pi}{6} \Rightarrow u = 1 + \sin\frac{\pi}{6} = 1 + \frac{1}{2} = \frac{3}{2}$$

Also,

$$\sin x = u - 1$$

so

$$\cos 2x = 1 - 2\sin^2 x = 1 - 2(u-1)^2$$

Now expand:

$$\begin{aligned}\cos 2x &= 1 - 2(u^2 - 2u + 1)\\ &= 1 - 2u^2 + 4u - 2\\ &= -2u^2 + 4u - 1\end{aligned}$$

So the integral becomes

$$\begin{aligned}I &= \int_1^{3/2} \frac{\cos 2x}{u^{3/2}}\,\mathrm{d}u\\ &= \int_1^{3/2} \frac{-2u^2 + 4u - 1}{u^{3/2}}\,\mathrm{d}u\\ &= \int_1^{3/2} \left(-2u^{1/2} + 4u^{-1/2} - u^{-3/2}\right)\,\mathrm{d}u\end{aligned}$$

Integrating term by term,

$$\begin{aligned}I &= \left[-2\cdot\frac{2}{3}u^{3/2} + 4\cdot 2u^{1/2} - \frac{u^{-1/2}}{-1/2}\right]_1^{3/2}\\ &= \left[-\frac{4}{3}u^{3/2} + 8u^{1/2} + 2u^{-1/2}\right]_1^{3/2}\end{aligned}$$

Now evaluate at $u = \frac{3}{2}$:

$$\left(\frac{3}{2}\right)^{1/2} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}$$
$$\left(\frac{3}{2}\right)^{3/2} = \frac{3}{2}\cdot\frac{\sqrt{6}}{2} = \frac{3\sqrt{6}}{4}$$

$$\left(\frac{3}{2}\right)^{-1/2} = \frac{1}{\sqrt{3/2}} = \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$$

Hence

$$-\frac{4}{3}\left(\frac{3\sqrt{6}}{4}\right) + 8\left(\frac{\sqrt{6}}{2}\right) + 2\left(\frac{\sqrt{6}}{3}\right) = -\sqrt{6} + 4\sqrt{6} + \frac{2\sqrt{6}}{3}$$
$$= \frac{11\sqrt{6}}{3}$$

At $u = 1$,

$$-\frac{4}{3}(1)^{3/2} + 8(1)^{1/2} + 2(1)^{-1/2} = -\frac{4}{3} + 8 + 2$$
$$= \frac{26}{3}$$

Therefore

$$I = \frac{11\sqrt{6}}{3} - \frac{26}{3}$$
$$= \frac{11\sqrt{6} - 26}{3}$$

So the exact value is

$$-\frac{26}{3} + \frac{11}{3}\sqrt{6}$$

Hence

$$a = -\frac{26}{3}, \qquad b = \frac{11}{3}$$

11. Use an appropriate substitution to find the exact value of

$$\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\pi$, where a and b are rational numbers. **[9]**

Solution

Use the substitution

$$x = 4\sin\theta$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 4\cos\theta \qquad \Rightarrow \qquad \mathrm{d}x = 4\cos\theta\,\mathrm{d}\theta$$

Also,

$$\begin{aligned}\sqrt{16-x^2} &= \sqrt{16-16\sin^2\theta}\\ &= \sqrt{16(1-\sin^2\theta)}\\ &= \sqrt{16\cos^2\theta}\\ &= 4\cos\theta\end{aligned}$$

Since the limits will give $0 \leq \theta \leq \frac{\pi}{4}$, we have $\cos\theta \geq 0$, so this square root is indeed $4\cos\theta$.
Now change the limits.
When $x = 0$,

$$0 = 4\sin\theta \qquad \Rightarrow \qquad \sin\theta = 0 \qquad \Rightarrow \qquad \theta = 0$$

When $x = 2\sqrt{2}$,

$$2\sqrt{2} = 4\sin\theta \qquad \Rightarrow \qquad \sin\theta = \frac{\sqrt{2}}{2} \qquad \Rightarrow \qquad \theta = \frac{\pi}{4}$$

So the integral becomes

$$\begin{aligned}\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}}\,\mathrm{d}x &= \int_0^{\pi/4} \frac{(4\sin\theta)^2}{4\cos\theta}\cdot 4\cos\theta\,\mathrm{d}\theta\\ &= \int_0^{\pi/4} \frac{16\sin^2\theta}{4\cos\theta}\cdot 4\cos\theta\,\mathrm{d}\theta\\ &= 16\int_0^{\pi/4} \sin^2\theta\,\mathrm{d}\theta\end{aligned}$$

Use the identity

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

Then

$$\begin{aligned}16\int_0^{\pi/4} \sin^2\theta\,\mathrm{d}\theta &= 16\int_0^{\pi/4} \frac{1-\cos 2\theta}{2}\,\mathrm{d}\theta\\ &= 8\int_0^{\pi/4} (1-\cos 2\theta)\,\mathrm{d}\theta\\ &= 8\left[\theta - \frac{1}{2}\sin 2\theta\right]_0^{\pi/4}\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}8\left[\theta - \frac{1}{2}\sin 2\theta\right]_0^{\pi/4} &= 8\left(\frac{\pi}{4} - \frac{1}{2}\sin\frac{\pi}{2}\right) - 8\left(0 - \frac{1}{2}\sin 0\right)\\ &= 8\left(\frac{\pi}{4} - \frac{1}{2}\right)\\ &= 2\pi - 4\end{aligned}$$

So in the form $a + b\pi$,

$$a = -4, \qquad b = 2$$

12. Find the exact value of

$$\int_0^{\frac{1}{2}\ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9\tanh^2 x}}\,\mathrm{d}x \qquad [8]$$

Solution

Let

$$I = \int_0^{\frac{1}{2}\ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9\tanh^2 x}}\,\mathrm{d}x$$

Use the substitution

$$u = \frac{3}{2}\tanh x$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{3}{2}\operatorname{sech}^2 x$$

so

$$\mathrm{d}u = \frac{3}{2}\operatorname{sech}^2 x\,\mathrm{d}x \qquad \Rightarrow \qquad \operatorname{sech}^2 x\,\mathrm{d}x = \frac{2}{3}\,\mathrm{d}u$$

Also, from $u = \frac{3}{2}\tanh x$,

$$\tanh x = \frac{2u}{3}$$

so

$$\begin{aligned}\sqrt{4-9\tanh^2 x} &= \sqrt{4-9\left(\frac{2u}{3}\right)^2}\\ &= \sqrt{4-4u^2}\\ &= 2\sqrt{1-u^2}\end{aligned}$$

Now change the limits.
When $x = 0$,

$$u = \frac{3}{2}\tanh 0 = 0$$

When $x = \frac{1}{2}\ln(2+\sqrt{3})$, we have

$$e^{2x} = 2+\sqrt{3}$$

and

$$\begin{aligned}\tanh x &= \frac{e^{2x}-1}{e^{2x}+1}\\ &= \frac{(2+\sqrt{3})-1}{(2+\sqrt{3})+1}\\ &= \frac{1+\sqrt{3}}{3+\sqrt{3}}\end{aligned}$$

Rationalising,

$$\begin{aligned}\frac{1+\sqrt{3}}{3+\sqrt{3}} &= \frac{(1+\sqrt{3})(3-\sqrt{3})}{(3+\sqrt{3})(3-\sqrt{3})}\\ &= \frac{3-\sqrt{3}+3\sqrt{3}-3}{9-3}\\ &= \frac{2\sqrt{3}}{6}\\ &= \frac{1}{\sqrt{3}}\end{aligned}$$

Hence

$$u = \frac{3}{2} \cdot \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2}$$

So the integral becomes

$$\begin{aligned} I &= \int_0^{\sqrt{3}/2} \frac{\frac{2}{3}\,\mathrm{d}u}{2\sqrt{1-u^2}} \\ &= \frac{1}{3}\int_0^{\sqrt{3}/2} \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u \end{aligned}$$

Now

$$\int \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u = \arcsin u$$

so

$$\begin{aligned} I &= \frac{1}{3}\left[\arcsin u\right]_0^{\sqrt{3}/2} \\ &= \frac{1}{3}\left(\arcsin\frac{\sqrt{3}}{2} - \arcsin 0\right) \\ &= \frac{1}{3}\left(\frac{\pi}{3} - 0\right) \\ &= \frac{\pi}{9} \end{aligned}$$

Therefore,

$$\int_0^{\frac{1}{2}\ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4 - 9\tanh^2 x}}\,\mathrm{d}x = \frac{\pi}{9}$$

13. Find the exact value of

$$\int_2^4 \frac{\sqrt{x^2-4}}{x^2}\,\mathrm{d}x$$

Give your answer in simplest exact form. **[9]**

Solution

Let

$$I = \int_2^4 \frac{\sqrt{x^2-4}}{x^2}\,\mathrm{d}x$$

Use the substitution

$$x = 2\sec\theta$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\sec\theta\tan\theta \quad \Rightarrow \quad \mathrm{d}x = 2\sec\theta\tan\theta\,\mathrm{d}\theta$$

Also,

$$\begin{aligned}\sqrt{x^2-4} &= \sqrt{4\sec^2\theta - 4}\\ &= \sqrt{4(\sec^2\theta - 1)}\\ &= \sqrt{4\tan^2\theta}\\ &= 2\tan\theta\end{aligned}$$

For the limits:
When $x = 2$,

$$2 = 2\sec\theta \Rightarrow \sec\theta = 1 \Rightarrow \theta = 0$$

When $x = 4$,

$$4 = 2\sec\theta \Rightarrow \sec\theta = 2 \Rightarrow \theta = \frac{\pi}{3}$$

So the integral becomes

$$\begin{aligned}I &= \int_0^{\pi/3} \frac{2\tan\theta}{(2\sec\theta)^2}\cdot 2\sec\theta\tan\theta\,\mathrm{d}\theta\\ &= \int_0^{\pi/3} \frac{2\tan\theta}{4\sec^2\theta}\cdot 2\sec\theta\tan\theta\,\mathrm{d}\theta\\ &= \int_0^{\pi/3} \frac{\tan^2\theta}{\sec\theta}\,\mathrm{d}\theta\end{aligned}$$

Now use $\tan^2\theta = \sec^2\theta - 1$:

$$\begin{aligned}\frac{\tan^2\theta}{\sec\theta} &= \frac{\sec^2\theta - 1}{\sec\theta}\\ &= \sec\theta - \frac{1}{\sec\theta}\\ &= \sec\theta - \cos\theta\end{aligned}$$

Hence

$$I = \int_0^{\pi/3} (\sec\theta - \cos\theta)\,\mathrm{d}\theta$$

Integrating,

$$\begin{aligned}I &= [\ln(\sec\theta + \tan\theta) - \sin\theta]_0^{\pi/3}\\ &= \left(\ln\left(2+\sqrt{3}\right) - \sin\frac{\pi}{3}\right) - (\ln(1+0) - \sin 0)\\ &= \ln\left(2+\sqrt{3}\right) - \frac{\sqrt{3}}{2}\end{aligned}$$

Alternative Solution:
Use the substitution

$$x = 2\cosh u$$

Then

$$\frac{\mathrm{d}x}{\mathrm{d}u} = 2\sinh u \quad \Rightarrow \quad \mathrm{d}x = 2\sinh u\,\mathrm{d}u$$

Also, since $u \geq 0$ on this interval,

$$\sqrt{x^2-4} = \sqrt{4\cosh^2 u - 4} = 2\sinh u$$

When $x = 2$, $\cosh u = 1$, so $u = 0$. When $x = 4$, $\cosh u = 2$, so $u = \operatorname{arcosh} 2 = \ln\left(2+\sqrt{3}\right)$.
So the integral becomes

$$\begin{aligned} I &= \int_0^{\ln(2+\sqrt{3})} \frac{2\sinh u}{4\cosh^2 u} \cdot 2\sinh u\,\mathrm{d}u \\ &= \int_0^{\ln(2+\sqrt{3})} \tanh^2 u\,\mathrm{d}u \\ &= \int_0^{\ln(2+\sqrt{3})} \left(1 - \operatorname{sech}^2 u\right)\mathrm{d}u \\ &= \left[u - \tanh u\right]_0^{\ln(2+\sqrt{3})} \end{aligned}$$

At the upper limit, $\cosh u = 2$ and $\sinh u = \sqrt{\cosh^2 u - 1} = \sqrt{3}$, so $\tanh u = \frac{\sqrt{3}}{2}$. Hence

$$I = \ln\left(2+\sqrt{3}\right) - \frac{\sqrt{3}}{2}$$

14. Show that

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2+\cos\theta}\,\mathrm{d}\theta = 15\ln\left(\frac{3}{2}\right) - 6 \qquad \textbf{[7]}$$

Solution

Let

$$I = \int_0^{\pi/2} \frac{\sin 3\theta}{2+\cos\theta}\,\mathrm{d}\theta$$

First rewrite $\sin 3\theta$ in terms of $\sin\theta$ and $\cos\theta$.
Using

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

and $\sin^2\theta = 1-\cos^2\theta$,

$$\begin{aligned}\sin 3\theta &= \sin\theta\big(3-4\sin^2\theta\big)\\ &= \sin\theta\big(3-4(1-\cos^2\theta)\big)\\ &= \sin\theta(4\cos^2\theta-1)\end{aligned}$$

So

$$I = \int_0^{\pi/2} \frac{\sin\theta(4\cos^2\theta-1)}{2+\cos\theta}\,\mathrm{d}\theta$$

Now use the substitution

$$u = \cos\theta \qquad \Rightarrow \qquad \mathrm{d}u = -\sin\theta\,\mathrm{d}\theta$$

When $\theta = 0$, $u = 1$.
When $\theta = \frac{\pi}{2}$, $u = 0$.
Hence

$$\begin{aligned}I &= \int_0^{\pi/2} \frac{\sin\theta(4\cos^2\theta-1)}{2+\cos\theta}\,\mathrm{d}\theta\\ &= \int_1^0 \frac{4u^2-1}{u+2}(-\mathrm{d}u)\\ &= \int_0^1 \frac{4u^2-1}{u+2}\,\mathrm{d}u\end{aligned}$$

Now divide $4u^2-1$ by $u+2$:

$$4u^2-1 = (u+2)(4u-8)+15$$

So

$$\frac{4u^2-1}{u+2} = 4u-8+\frac{15}{u+2}$$

Therefore

$$\begin{aligned}I &= \int_0^1 \left(4u-8+\frac{15}{u+2}\right)\mathrm{d}u\\ &= \big[2u^2-8u+15\ln(u+2)\big]_0^1\end{aligned}$$

Now substitute the limits:

$$\begin{aligned}I &= \big(2(1)^2-8(1)+15\ln 3\big)-\big(2(0)^2-8(0)+15\ln 2\big)\\ &= (2-8+15\ln 3)-15\ln 2\\ &= -6+15\ln\left(\frac{3}{2}\right)\end{aligned}$$

Hence

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2+\cos\theta}\,\mathrm{d}\theta = 15\ln\left(\frac{3}{2}\right) - 6$$

So the required result is shown.

15. Show that

$$\int_0^{\pi/2} \frac{\cos^5\theta}{2+\sin\theta}\,\mathrm{d}\theta = 9\ln\left(\frac{3}{2}\right) - \frac{41}{12} \qquad \textbf{[8]}$$

Solution

Let

$$I = \int_0^{\pi/2} \frac{\cos^5\theta}{2+\sin\theta}\,\mathrm{d}\theta$$

Use the substitution

$$u = \sin\theta$$

so that

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = \cos\theta \qquad \Rightarrow \qquad \mathrm{d}u = \cos\theta\,\mathrm{d}\theta$$

Also,

$$\cos^5\theta = \cos^4\theta\cos\theta = (1-\sin^2\theta)^2\cos\theta = (1-u^2)^2\cos\theta$$

When $\theta = 0$, $u = \sin 0 = 0$, and when $\theta = \frac{\pi}{2}$, $u = \sin\frac{\pi}{2} = 1$.
So the integral becomes

$$\begin{aligned} I &= \int_0^1 \frac{(1-u^2)^2}{2+u}\,\mathrm{d}u \\ &= \int_0^1 \frac{1-2u^2+u^4}{u+2}\,\mathrm{d}u \end{aligned}$$

Now divide $u^4 - 2u^2 + 1$ by $u + 2$:

$$\frac{u^4-2u^2+1}{u+2} = u^3 - 2u^2 + 2u - 4 + \frac{9}{u+2}$$

Hence

$$I = \int_0^1 \left(u^3 - 2u^2 + 2u - 4 + \frac{9}{u+2}\right)\,\mathrm{d}u$$

Integrating term by term,

$$I = \left[\frac{u^4}{4} - \frac{2u^3}{3} + u^2 - 4u + 9\ln(u+2)\right]_0^1$$

Now substitute the limits:

$$\begin{aligned} I &= \left(\frac{1}{4} - \frac{2}{3} + 1 - 4 + 9\ln 3\right) - (0 - 0 + 0 - 0 + 9\ln 2) \\ &= \frac{1}{4} - \frac{2}{3} + 1 - 4 + 9(\ln 3 - \ln 2) \\ &= \frac{3 - 8 + 12 - 48}{12} + 9\ln\left(\frac{3}{2}\right) \\ &= -\frac{41}{12} + 9\ln\left(\frac{3}{2}\right) \end{aligned}$$

Therefore,

$$\int_0^{\pi/2} \frac{\cos^5\theta}{2+\sin\theta}\,\mathrm{d}\theta = 9\ln\left(\frac{3}{2}\right) - \frac{41}{12}$$

as required.